

MooreVIEW Lift Station + EZ Meter — Parc & Cloud Commissioning Guide

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Audience: Panel builders, field technicians, cloud commissioning engineers
Firmware: `MooreviewOptaMqttSt` with `-DMV_FIELDBUS=1 -DMV_EZMETER=1`

Purpose

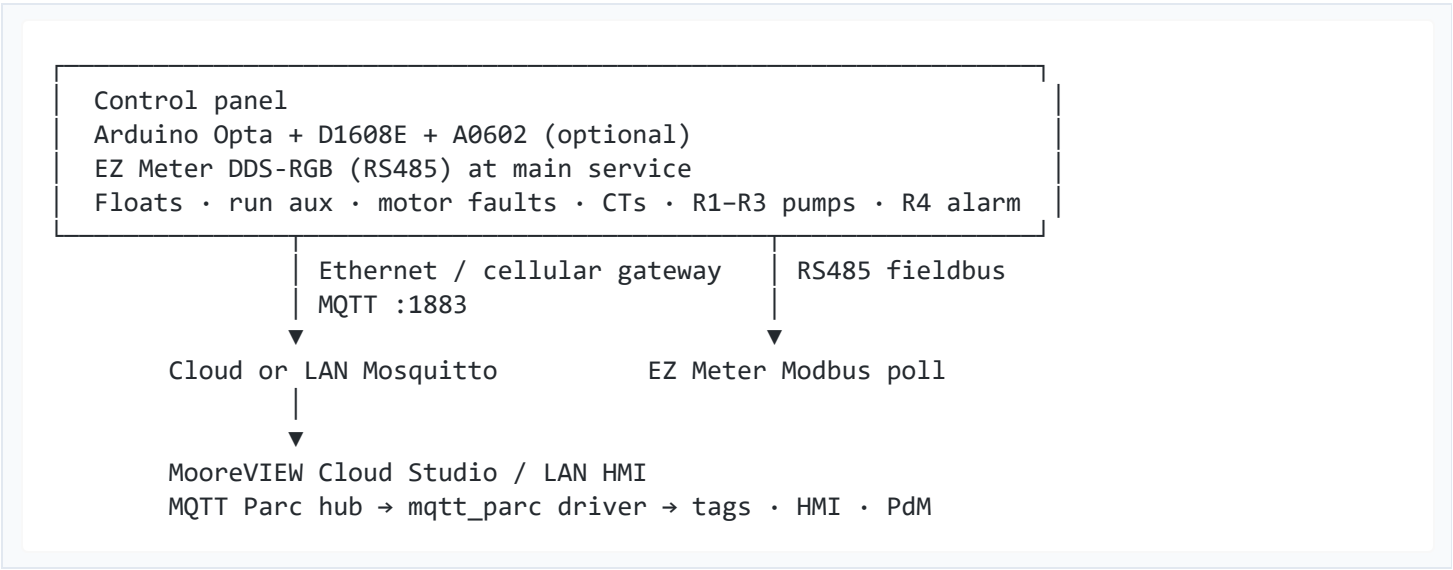
This guide covers **building a lift-station control panel, wiring Opta + Sequent expansion + EZ Meter**, and **configuring MooreVIEW Parc (MQTT) and Cloud Studio** for hardware bench testing and field deployment.

Two station types are supported:

Type	ST program	Cloud template	Pumps	Floats
Duplex	<code>logic/38_duplex_lift_station_ezmeter.st</code>	<code>lift_station_dual_duplex_ezmeter</code>	2 (R1, R2)	4 (HIGH, LEAD, LAG, OFF)
Triplex	<code>logic/39_triplex_lift_station_ezmeter.st</code>	<code>lift_station_triplex_ezmeter</code>	3 (R1–R3)	5 (+ LAG2)

Both use **R4** as the **station alarm relay** (horn, strobe, or SCADA alarm contact).

1. Architecture



Direct to cloud: Opta broker = `mooreview.io` or droplet IP, port **1883** (plain MQTT — no TLS on Opta yet).

Via gateway: Opta broker = `192.168.1.1` (LilyGO cellular gateway LAN).

See [OPTA PARC CLOUD.md](#) and [CSTORE OPTA PARC CLOUD.md](#).

2. Panel bill of materials

Qty	Item	Notes
1	Arduino Opta (AFX00001 / WiFi variant optional)	Base I/O, RS485, R1–R4 relays
1	Sequent D1608E (AFX00005)	Slot 1 — 16 DI + 8 relay (use DIs; pump relays on Opta base)
1	Sequent A0602 (AFX00007)	Slot 2 optional — RTD / spare analog
1	EZ Meter DDS-RGB (or DDS-RGB 2.025)	Polyphase meter, Modbus RTU, RS485
2–3	0–1 V CT transmitters (per pump, 3-phase)	1.00 V = 50 A linear
2–3	Pump contactors	Driven by R1, R2 (duplex) or R1–R3 (triplex)
1	Alarm horn/strobe or dry contact	Driven by R4
4–5	Float switches	Pull-down ladder (+24 V when wet)
2–3	Motor fault inputs	Overload / seal failure contacts → expansion DIs
1	24 VDC power supply	Opta + expansion + field devices
1	Ethernet or cellular gateway	MQTT path to MooreVIEW

3. Opta base unit — relay outputs

Pump contactors and station alarm are wired to the **Opta base** relay terminals (not expansion relays).

Terminal	Channel	Type	ST tag	Function
R1	R1	BOOL	MOTOR1_RUN	Pump 1 contactor
R2	R2	BOOL	MOTOR2_RUN	Pump 2 contactor
R3	R3	BOOL	— / MOTOR3_RUN	Spare (duplex) · Pump 3 contactor (triplex)
R4	R4	BOOL	R4	Station alarm (high level / EZ Meter PQ / motor fault)

R4 energizes when any of:

- **High level float** — LVL_HIGH (X1_I1)
- **EZ Meter PQ alarm** — MECH_PQ_ALM (undervolt, overvolt, phase loss, imbalance, low PF, frequency)
- **Motor fault DIs** — see expansion I/O map below

Phase fault (X1_I7 duplex / X1_I9 triplex) **interlocks auto pump demand** but does **not** drive R4.

4. Opta base unit — digital inputs

Opta terminals **I1–I6** are configured as **analog** CT inputs (see §5). Terminals **I7** and **I8** remain available as digital inputs.

Terminal	Channel	Type	ST tag	Function
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I7	I7	BOOL	—	Spare digital (I1–I6 used as analog CT inputs)
I8	I8	BOOL	—	Spare digital

Wiring: +24 V active when contact made (same convention as expansion DIs). Unwired inputs read OFF.

5. Opta base unit — analog inputs (motor CTs)

Motor current is measured on the **Opta base** using **0–1 V CT transmitters** (one per phase per pump). Terminals **I1–I6** are the physical analog inputs; firmware publishes raw counts as **I1_RAW ... I6_RAW** and engineering amps as **AI1 ... AI6**.

Terminal	Channel (raw)	Engineering tag	Function	Scale
AI1 / I1	I1_RAW	AI1	Motor 1 phase A amps	0–1 V → 0–50 A (1.00 V = 50 A)
AI2 / I2	I2_RAW	AI2	Motor 1 phase B amps	Engineering = raw × 0.48828125
AI3 / I3	I3_RAW	AI3	Motor 1 phase C amps	
AI4 / I4	I4_RAW	AI4	Motor 2 phase A amps	
AI5 / I5	I5_RAW	AI5	Motor 2 phase B amps	
AI6 / I6	I6_RAW	AI6	Motor 2 phase C amps	
I7	I7_RAW	—	Spare raw (if I7 configured analog)	
I8	I8_RAW	—	Spare raw (if I8 configured analog)	

Wiring notes:

- CT transmitters: linear **0–1 VDC**, **1.00 V = 50 A** full scale.
 - Duplex: AI1–AI3 = pump 1 (A/B/C); AI4–AI6 = pump 2 (A/B/C).
 - Triplex: same AI1–AI6 for pumps 1 and 2; pump 3 has no base CT inputs in program 39 (PdM uses run status / optional **MOTOR3_TEMP** on expansion).
 - Bench test: apply **1.0 VDC** to AI1 → expect ~**50 A** on tag **AI1** (after CT calibration if enabled in Tags).
- These base analog inputs feed **PdM / MCSA** on the cloud host — separate from EZ Meter **DDS_I_*** at the main service.

6. Duplex expansion I/O (D1608E slot 1)

Expansion DIs: **+24 V active** (input ON when float wet or contact made).

Terminal	Tag	Function
X1_I1	LVL_HIGH	High level float
X1_I2	LVL_LEAD	Lead float
X1_I3	LVL_LAG	Lag float

X1_I4	LVL_OFF	Off float
X1_I5	P1_RUN_FB	Pump 1 run aux (fail-to-run)
X1_I6	P2_RUN_FB	Pump 2 run aux (fail-to-run)
X1_I7	PHASE_FAULT	Phase loss / monitor relay
X1_I8	GEN_RUN	Generator running
X1_I9	GEN_FUEL_FAULT	Generator fuel fault
X1_I10	GEN_FAULT	Generator fault
X1_I11	MOTOR1_FAULT	Pump 1 motor fault → R4
X1_I12	MOTOR2_FAULT	Pump 2 motor fault → R4
X1_I13–I16	—	Spare

Expansion relays (slot 1, spare): X1_R1 ... X1_R8 — not used (pump contactors on base R1/R2).

7. Triplex expansion I/O (D1608E slot 1)

Terminal	Tag	Function
X1_I1	LVL_HIGH	High level float
X1_I2	LVL_LEAD	Lead float
X1_I3	LVL_LAG	Lag float
X1_I4	LVL_LAG2	Lag2 float (3-pump stage)
X1_I5	LVL_OFF	Off float
X1_I6	P1_RUN_FB	Pump 1 run aux
X1_I7	P2_RUN_FB	Pump 2 run aux
X1_I8	P3_RUN_FB	Pump 3 run aux
X1_I9	PHASE_FAULT	Phase fault
X1_I10	GEN_RUN	Generator running
X1_I11	GEN_FUEL_FAULT	Gen fuel fault
X1_I12	GEN_FAULT	Generator fault
X1_I13	MOTOR1_FAULT	Pump 1 motor fault → R4
X1_I14	MOTOR2_FAULT	Pump 2 motor fault → R4
X1_I15	MOTOR3_FAULT	Pump 3 motor fault → R4
X1_I16	—	Spare

Relays: R1–R3 = pumps (base unit); **R4** = station alarm.

Regenerate triplex ST/fixture after duplex changes: `node scripts/lift-station/generate-triplex-ezmeter.js` .

8. Expansion slot 2 — A0602 analog (optional)

Sequent **A0602** in expansion slot 2 provides RTD / analog inputs for motor winding temperature and spare channels.

Terminal	Channel	Engineering tag	Function
X2_AI1	X2_AI1	MOTOR1_TEMP	Motor 1 winding temp (calibrate in Tags)
X2_AI2	X2_AI2	MOTOR2_TEMP	Motor 2 winding temp (triplex: also pump 3 temp binding)
X2_AI3–AI8	X2_AI3 ... X2_AI8	—	Spare analog
X2_PWM1–PWM4	X2_PWM1 ... X2_PWM4	—	Spare PWM outputs

Disable slot 2 on Opta `/setup` if the A0602 module is not installed.

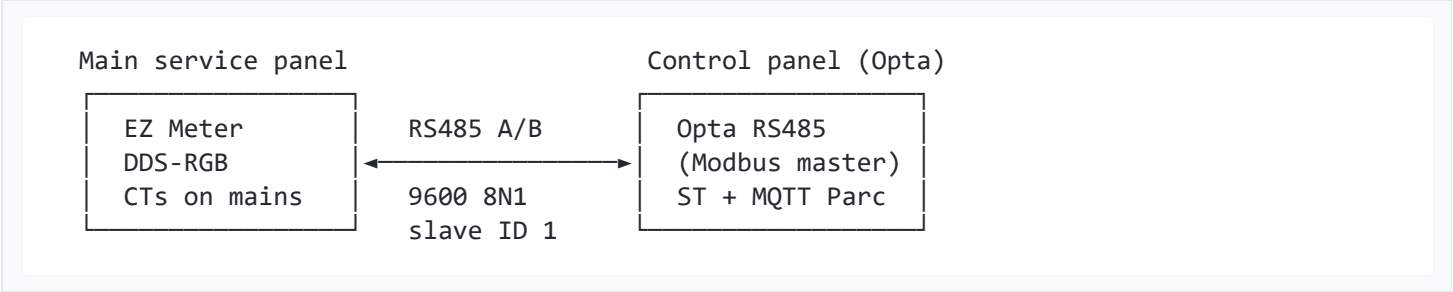
Complete I/O map PDF: `npm run build:duplex-io-map-pdf` → `docs/projects/duplex-lift-station-opta-io-map.pdf` (base unit + all expansion sections).

9. EZ Meter — wiring, Modbus, and tag data

The lift-station programs embed **EZ Meter facility PQ** inline (no separate `ezmeter_facility_pq.st` deploy). Opta firmware polls the meter on RS485; ST computes PQ alarms and drives **R4** via `MECH_PQ_ALM` .

9.1 Physical wiring

Item	Detail
Meter model	EZ Meter DDS-RGB 2.025 (RGB firmware v1.600)
Location	Main service / mechanical room panel (upstream of pump loads)
Opta port	Onboard RS485 terminal block (A / B / GND)
Cable	Twisted pair, shield optional; keep away from pump VFD output wiring
Termination	120 Ω at far end if bus length > 10 m (meter may have onboard term jumper)
Power	Meter line voltage per nameplate (typically 120/240 V or 208/480 V service)
CTs	Meter CTs on main service conductors (separate from pump CT inputs on Opta AI1–AI6)



Confirm slave ID and baud on the meter display or EZ Meter configuration tool before commissioning.

9.2 Modbus RTU parameters

Parameter	Default	Notes
Protocol	Modbus RTU	Opta = master, meter = slave
Baud	9600	8 data bits, no parity, 1 stop (8N1)
Slave ID	1	Override in template <code>fieldbus.ezmeter.slaveId</code>
Live poll	30 s	<code>MV_EZMETER_LIVE_POLL_MS</code> — V, I, W, Hz, PF, VA
Energy poll	5 min	<code>MV_EZMETER_ENERGY_POLL_MS</code> — summed Wh import/export
Register type	Holding (FC 03)	HR 40001+ metered values; HR 41001+ control/status

Firmware log on success: `[EZM] RS485 EZ Meter poll slave=1` . Poll status tag: `EZM_POLL_STA` (`OK` / `poll_err`).

9.3 Modbus register map — live analog (HR 40025–40047)

Raw register × scale = engineering units. Opta firmware writes these to `DDS_*` tags every live poll.

HR	Tag	Type	Scale	Units	Description
40025	DDS_V_A	REAL	×0.1	V	Phase A voltage
40026	DDS_I_A	REAL	×0.1	A	Phase A current
40027–28	DDS_W_A	REAL	×0.1	W	Phase A real power (32-bit signed)
40029	DDS_HZ_A	REAL	×0.1	Hz	Phase A frequency
40030	DDS_PF_A	REAL	×0.01	—	Phase A power factor
40031	DDS_V_B	REAL	×0.1	V	Phase B voltage
40032	DDS_I_B	REAL	×0.1	A	Phase B current
40033–34	DDS_W_B	REAL	×0.1	W	Phase B real power
40035	DDS_HZ_B	REAL	×0.1	Hz	Phase B frequency
40036	DDS_PF_B	REAL	×0.01	—	Phase B power factor
40037	DDS_V_C	REAL	×0.1	V	Phase C voltage

40038	DDS_I_C	REAL	×0.1	A	Phase C current
40039–40	DDS_W_C	REAL	×0.1	W	Phase C real power
40041	DDS_HZ_C	REAL	×0.1	Hz	Phase C frequency
40042	DDS_PF_C	REAL	×0.01	—	Phase C power factor
40043–44	DDS_VA_A	REAL	×0.1	VA	Phase A apparent power
40045–46	DDS_VA_B	REAL	×0.1	VA	Phase B apparent power
40047–48	DDS_VA_C	REAL	×0.1	VA	Phase C apparent power

9.4 Modbus register map — energy (HR 40001–40023)

Polled every **5 minutes**. Wh registers are 32-bit; raw × 10 = Wh.

HR	Tag	Scale	Units	Description
40001–02	DDS_WH_A_IMP	×10	Wh	Phase A import energy
40003–04	DDS_WH_B_IMP	×10	Wh	Phase B import energy
40005–06	DDS_WH_C_IMP	×10	Wh	Phase C import energy
40009–10	DDS_WH_A_EXP	×10	Wh	Phase A export energy
40011–12	DDS_WH_B_EXP	×10	Wh	Phase B export energy
40013–14	DDS_WH_C_EXP	×10	Wh	Phase C export energy
40017–18	DDS_WH_SUM_IMP	×10	Wh	Summed import Wh (primary kWh source)
40019–20	DDS_WH_SUM_EXP	×10	Wh	Summed export Wh
40021–22	DDS_VAH_SUM_IMP	×10	VAh	Summed apparent import
40023–24	DDS_VAH_SUM_EXP	×10	VAh	Summed apparent export

kWh conversion: `MECH_METER_KWH` = `DDS_WH_SUM_IMP` × 0.001 (Wh → kWh after ×10 scale).

Per-phase Wh tags (`DDS_WH_A_IMP` ...) are in the full Modbus map template but are **not** polled by Opta firmware — only live analog + summed Wh on the edge.

9.5 Modbus register map — control / status (HR 41011–41034)

Read-only identification registers. Available in cloud Modbus template; not polled on Opta edge.

HR	Tag	Description
41011	DDS_CTL_PROT_LVL	Protection level
41012–13	DDS_CTL_SER_NO	Serial number
41014	DDS_CTL_DATA_BASE	Data register base (typically 40001)
41015	DDS_CTL_CTL_BASE	Control register base (typically 41001)

41016	DDS_CTL_COMM	Comm settings (0x13 = 9600 8N1)
41018–22	DDS_CTL_MODEL_W1 ... W5	Model string (ASCII)
41024–33	DDS_CTL_CUST_W1 ... W10	Customer ID (ASCII)

Source: `src/facilities/ezmeterRegisterMap.js` · extract: `st/fixtures/dds-rgb-modbus-extract.txt` .

9.6 Tags published by Opta firmware (MQTT Parc)

These tags are updated on-device and published via MQTT telemetry:

Tag group	Tags	Update rate
Live analog	DDS_V_A/B/C, DDS_I_A/B/C, DDS_W_A/B/C, DDS_HZ_A/B/C, DDS_PF_A/B/C, DDS_VA_A/B/C	~30 s
Energy	DDS_WH_SUM_IMP, DDS_WH_SUM_EXP	~5 min
THD estimate	MECH_PQ_THD_VA, MECH_PQ_THD_VB, MECH_PQ_THD_VC, MECH_PQ_THD_IA, MECH_PQ_THD_IB, MECH_PQ_THD_IC	~30 s (PF-derived)
THD alarm	MECH_PQ_THD_ALM, MECH_PQ_CFG_THD_PCT	~30 s
PQ config (NV)	MECH_PQ_CFG_NOM_V, MECH_PQ_CFG_UV_V	On boot + cloud push

THD note: RGB v1.600 has **no harmonic registers**. Firmware estimates THD from power factor (`MECH_PQ_THD_*`). Treat as indicative only — not revenue-grade harmonic analysis.

9.7 Derived PQ tags (ST program 38 / 39)

Computed each ST scan from `DDS_*` inputs. Visible on **Tags** → **Live** after program is running.

Tag	Type	Formula / source
MECH_PQ_KW_SUM	REAL	(DDS_W_A + DDS_W_B + DDS_W_C) / 1000 kW
MECH_PQ_KVA_SUM	REAL	(DDS_VA_A + DDS_VA_B + DDS_VA_C) / 1000 kVA
MECH_PQ_PF_SYS	REAL	MECH_PQ_KW_SUM / MECH_PQ_KVA_SUM
MECH_PQ_V_AVG	REAL	Average of DDS_V_A/B/C
MECH_PQ_V_IMBAL_PCT	REAL	Max–min phase V / average × 100
MECH_PQ_I_SUM	REAL	DDS_I_A + DDS_I_B + DDS_I_C
MECH_METER_INTERVAL_KWH	REAL	Delta kWh since last scan
DDS_METER_KWH_PREV	REAL	Previous kWh snapshot (internal)

9.8 Facility mirror tags (semantic map)

These tags mirror `DDS_*` for HMI bindings and campus dashboards:

Facility tag	Source tag	Scale	Units
MECH_METER_KWH	DDS_WH_SUM_IMP	×0.001	kWh
MECH_METER_KWH_EXP	DDS_WH_SUM_EXP	×0.001	kWh
MECH_PQ_VA	DDS_V_A	1	V
MECH_PQ_VB	DDS_V_B	1	V
MECH_PQ_VC	DDS_V_C	1	V
MECH_PQ_IA	DDS_I_A	1	A
MECH_PQ_IB	DDS_I_B	1	A
MECH_PQ_IC	DDS_I_C	1	A
MECH_PQ_HZ	DDS_HZ_A	1	Hz
MECH_PQ_PF_A/B/C	DDS_PF_A/B/C	1	—

9.9 PQ alarm tags and R4 behavior

Alarm tag	Default condition	Effect on lift station
MECH_PQ_UNDERVOLT	Any phase V < 108 V	R4 ON ; blocks LEAD_CALL / LAG_CALL
MECH_PQ_OVERVOLT	Any phase V > 132 V	R4 ON
MECH_PQ_PHASE_LOSS	One phase < UV while others live	R4 ON ; blocks pump auto demand
MECH_PQ_V_IMBAL	Imbalance > 5%	R4 ON
MECH_PQ_LOW_PF	System PF low under load (> 1 A total)	R4 ON
MECH_PQ_FREQ_FAULT	Frequency outside 59.5–60.5 Hz	R4 ON
MECH_PQ_THD_ALM	Estimated THD > 8%	Alarm only (does not block pumps)
MECH_PQ_ALM	Any PQ alarm above	R4 ON (OR with LVL_HIGH , motor faults)

Undervolt and phase loss run **before** level demand in the ST scan so pump interlock matches phase-fault behavior.

9.10 Threshold configuration tags (MECH_PQ_CFG_*)

Tag	Default	Cloud setting key	Opta NV
MECH_PQ_CFG_NOM_V	120 V	Nominal voltage	✓ /setup
MECH_PQ_CFG_UV_V	108 V	Undervolt threshold	✓ /setup
MECH_PQ_CFG_OV_V	132 V	Overvolt threshold	Cloud only
MECH_PQ_CFG_LOW_PF	0.85	Low PF threshold	Cloud only
MECH_PQ_CFG_FREQ_MIN	59.5 Hz	Min frequency	Cloud only

MECH_PQ_CFG_FREQ_MAX	60.5 Hz	Max frequency	Cloud only
MECH_PQ_CFG_IMBAL_PCT	5%	Max voltage imbalance	Cloud only
MECH_PQ_CFG_LOAD_I	1 A	Loaded current threshold	Cloud only
MECH_PQ_CFG_THD_PCT	8%	THD alarm threshold	Firmware NV

Where to set:

Where	Path
Cloud	System setup → MQTT Parc → EZ Meter PQ
Opta	http://<opta-ip>/setup → EZ Meter PQ card (nominal + undervolt)

Saving cloud settings calls `syncEzMeterThresholdTags()` and pushes `MECH_PQ_CFG_*` to connected Opta drivers.

Settings JSON (project or tenant):

```

"mqttParc": {
  "ezMeter": {
    "nominalVoltage": 120,
    "undervoltV": 108,
    "overvoltV": 132,
    "lowPf": 0.85,
    "freqMinHz": 59.5,
    "freqMaxHz": 60.5,
    "vImbalancePct": 5,
    "loadedCurrentA": 1,
    "thdAlarmPct": 8
  }
}

```

9.11 MQTT telemetry path

Opta publishes all `DDS_*` and `MECH_PQ_*` tags on the standard Parc topic:

```
mooreview/v1/{deviceId}/telemetry
```

Payload includes tag id, value, type, and timestamp. Cloud **mqtt_parc** hub ingests telemetry → tag store → HMI composites and historian (when Hist enabled).

Enable historian on key PQ tags for post-event review: `DDS_V_*` , `DDS_PF_*` , `MECH_PQ_KW_SUM` , `MECH_PQ_V_IMBAL_PCT` , `MECH_PQ_ALM` .

9.12 Bench test — expected EZ Meter values

Condition	Expected tags	Pass
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Meter powered, healthy 120 V service	DDS_V_A/B/C $\approx$ 118–122 V	✓
No load	DDS_I_A/B/C $\approx$ 0–2 A (meter self-load)	✓
Line frequency	DDS_HZ_A $\approx$ 59.9–60.1 Hz	✓
Power factor	DDS_PF_A/B/C $\approx$ 0.95–1.0 (light load)	✓
Energy accumulating	DDS_WH_SUM_IMP increasing every 5 min	✓
Undervolt test	Temporarily set MECH_PQ_CFG_UV_V = 130 V	MECH_PQ_UNDERVOLT, MECH_PQ_ALM, R4 ON
Restore threshold	Set undervolt back to 108 V	Alarms clear, R4 off (if no other alarm)
RS485 fault	Disconnect A/B	EZM_POLL_STA = poll_err; DDS_V_* stale

See also [EZMETER FACILITY PQ.md](#) for standalone Modbus template apply and campus sync.

10. Firmware flash (Opta)

1. Open `firmware/arduino-opta-mqtt-st/MooreviewOptaMqttSt` in Arduino IDE.
2. **Tools** → **Board** → **Arduino Opta** (`mbed_opta`).
3. **Tools** → **Compile flags** (or `platform.local.txt`):

```
-DMV_FIELDBUS=1 -DMV_EZMETER=1
```

4. Install libraries: ArduinoJson 7.x, PubSubClient, Arduino_Opta_Blueprint, ArduinoModbus, ArduinoRS485.
5. Upload via USB/Ethernet.

Verify: Serial log shows `[EZM] RS485 EZ Meter poll slave=1` after boot.

11. Opta field configuration (/setup)

Open `http://<opta-ip>/setup` (or WiFi AP `MooreVIEW-Opta` → `http://192.168.4.1:8080/setup`).

11.1 Ethernet & expansion

Setting	Bench	Field
DHCP	✓ on LAN	As site requires
Expansion slot 1	AFX00005 D1608E	Scan expansions after save
Expansion slot 2	AFX00007 A0602 or Disabled	Optional RTD

11.2 MQTT Parc broker

Field	LAN bench	Cloud
Broker host	MooreVIEW PC IP or 127.0.0.1 via gateway	mooreview.io or droplet IP
Port	1883	1883
Username / password	Match Mosquitto	Match MOSQUITTO_USER / MOSQUITTO_PASS (≤ 47 chars)
Global site key	1 (0x0001)	Must match Cloud Studio

Click **Test MQTT connection** → expect success + `setup-test` publish.

11.3 EZ Meter PQ (Opta NV)

Field	Default
Nominal voltage	120 V
Undervolt threshold	108 V

Save settings → **Reboot device**.

Note **Device ID** on `/setup` (e.g. `mv_f2e689fd60d96bab`).

12. Cloud Studio configuration

Sign in → tenant → open or import project.

12.1 System setup → MQTT Parc

Setting	Value
Enable MQTT Parc hub	✓
Broker URL	mqtt://127.0.0.1:1883 (on cloud VM)
Username / password	Same as Mosquitto
Global site key	0x0001
Remote ST execution	✓
Auto-run ST on Opta after boot	✓ (after first deploy)

12.2 EZ Meter PQ (cloud)

Under **MQTT Parc** → **EZ Meter PQ**:

Field	Default
Nominal voltage (V)	120
Undervolt threshold (V)	108

Overvolt threshold (V)	132
THD alarm (%)	8

Saving pushes `MECH_PQ_CFG_*` tags to connected Opta drivers.

12.3 Add Opta driver

Drivers → **Add Opta Parc devices** (from registry):

Field	Example
Device ID	From Opta <code>/setup</code>
Position ID	Stable site slug, e.g. <code>bench-duplex-01</code>
Template	DUPLEXLS + EZ Meter or TRIPLEXLS + EZ Meter

Steps:

1. Wait for Parc registry → device **online**
2. **Sync tags from device**
3. **Scan expansions**
4. **Program** → **Remote** → **Connect** → **Download & Start**
 - Duplex: `logic/38_duplex_lift_station_ezmeter.st`
 - Triplex: `logic/39_triplex_lift_station_ezmeter.st`
5. Add HMI: **Composites** → **DUPLEXLS** or **TRIPLEXLS**

12.4 Generate bench project (local)

```
npm run generate:duplex-lift-station
```

Import `data/projects/duplex-lift-station.est.zip` for a pre-wired duplex demo.

13. Hardware bench test procedure

Use this checklist before shipping the panel or leaving the site.

13.1 Power and comms

Step	Pass criteria
24 VDC to Opta + expansion	Green power; expansion detected on <code>/setup</code>
Ethernet or gateway	Opta gets IP; pingable
MQTT test on <code>/setup</code>	Connected; <code>setup-test</code> on broker
Cloud registry	Device online, telemetry age < 30 s

13.2 Digital inputs (simulate with +24 V)

Test	Action	Expected tag
High float	Jumper X1_I1	LVL_HIGH ON; R4 ON
Lead float	Jumper X1_I2	LVL_LEAD ON; alternator demand
Motor fault	Jumper X1_I11 (duplex)	MOTOR1_FAULT ON; R4 ON
Phase fault	Jumper X1_I7 (duplex)	PHASE_FAULT ON; pumps interlocked; R4 off
Off float dry	Remove X1_I4 wet	Pumps de-stage

Verify on **Tags** → **Live I/O** or Opta `/io-map` .

13.3 Relay outputs

Test	Action	Expected
Pump 1	Hand mode → START (HMI) or force MOTOR1_RUN	R1 clicks
Pump 2	Same for pump 2	R2 clicks
Alarm	High float or undervolt	R4 clicks

13.4 EZ Meter / PQ

See **§9.12** for expected values. Quick checks:

Test	Action	Expected
Meter live	Apply line voltage to EZ Meter	DDS_V_A/B/C $\approx$ 118–122 V; DDS_HZ_A $\approx$ 60 Hz
Energy	Wait 5 min	DDS_WH_SUM_IMP increments; MECH_METER_KWH updates
kW sum	Run pumps under load	MECH_PQ_KW_SUM > 0; MECH_PQ_PF_SYS reasonable
Undervolt	Lower threshold in cloud to 130 V (temp)	MECH_PQ_UNDERVOLT, MECH_PQ_ALM, R4 ON ; pumps interlocked
Restore	Set undervolt back to 108 V	Alarms clear when voltage healthy
RS485	Serial log / EZM_POLL_STA	[EZM] RS485 EZ Meter poll slave=1; status OK

13.5 ST runtime

Check	Pass
Program → Remote → status	Running; scan OK
STATION_STA	0=Online when healthy
Fail-to-run	Run R1 without run aux → M1_FAIL after 5 s
HMI composite	Floats, pumps, alarms animate

13.6 CT inputs (base unit analog)

Apply **1.0 VDC** to **AI1** (terminal I1) → expect ~**50 A** on tag **AI1** . Repeat for AI2–AI6 per pump phase wiring (§5).

14. Cloud broker (production droplet)

On the cloud VM:

```
grep MOSQUITTO /etc/mooreview/env
sudo ufw allow 1883/tcp
mosquitto_sub -h 127.0.0.1 -p 1883 -u mooreview -P "$MOSQUITTO_PASS" \
  -t 'mooreview/v1/+/telemetry' -v
```

Env template: [deploy/cloud/.env.cstore-opta-parc.example](#).

Enable runtime on same droplet:

```
sudo MOOREVIEW_INSTALL_DIR=/home/mooreview bash deploy/cloud/debian/enable-runtime-3090.sh
```

15. Troubleshooting

Symptom	Cause	Fix
No expansion DIs	24 V missing on D1608E	Check expansion supply; Scan expansions on /setup
R4 always ON	Default DDS_V = 0 triggers undervolt	Set healthy meter voltage or raise undervolt threshold; set DDS_HZ_A path needs meter online
R4 never ON	Alarm relay wiring	Confirm R4 contactor; force LVL_HIGH
No DDS_* tags	RS485 or compile flags	Reflash with MV_FIELDBUS=1 MV_EZMETER=1 ; check A/B polarity
MQTT auth fail	Password > 47 chars	Shorten MOSQUITTO_PASS
Deploy timeout	Old firmware / hub off	Reflash v2.3.41+; enable MQTT Parc hub
LEAD_CALL stuck	PQ fault active	Clear undervolt; check EZ Meter wiring

MooreVIEW hints on Connect/Download failures: match broker host, auth, and firewall **1883**.

16. Reference files

Path	Purpose
------	---------

st/logic/38_duplex_lift_station_ezmeter.st	Duplex ST + EZ Meter PQ
st/logic/39_triplex_lift_station_ezmeter.st	Triplex ST + EZ Meter PQ
st/fixtures/tags.duplex_lift_station_ezmeter.json	Duplex tag fixture
st/fixtures/tags.triplex_lift_station_ezmeter.json	Triplex tag fixture
src/devices/templates/lift_station_dual_duplex_ezmeter.json	Duplex cloud template
src/devices/templates/lift_station_triplex_ezmeter.json	Triplex cloud template
src/devices/templates/arduino_opta_parc_ezmeter.json	Opta-only Parc + EZ Meter
docs/OPTA_PARC_CLOUD.md	Opta → cloud MQTT
docs/facilities/EZMETER_FACILITY_PQ.md	PQ tags and Modbus map
firmware/arduino-opta-mqtt-st/OPTa_FEATURES.md	Firmware features
scripts/duplex-lift-station/opta-io-map-data.js	I/O map source for PDF

PDF outputs:

```

npm run build:lift-station-ezmeter-parc-pdf    # This guide
npm run build:duplex-io-map-pdf               # Duplex I/O map only

```

17. Quick commissioning checklist

#	Task	✓
1	Panel wired per I/O map (§3–§8); R4 → alarm device; AI1–AI6 CT transmitters on base I1–I6	
2	EZ Meter on RS485; line voltage applied; DDS_v_* live	
3	Opta flashed with MV_FIELDBUS + MV_EZMETER	
4	/setup — broker, site key, EZ Meter undervolt saved	
5	MQTT test OK; device ID recorded	
6	Cloud — MQTT Parc hub + EZ Meter PQ settings	
7	Driver added; tags synced; expansions scanned	
8	ST 38 or 39 deployed Download & Start	
9	Bench floats, motor faults, R4 alarm verified; base CT inputs AI1–AI6 live (§5); EZ Meter PQ tags live (§9)	
10	HMI composite bound; PdM assets seeded	